Decision-Making Under Uncertainty

A Student Workbook of Four Decisions

One formula, applied four times. Two decisions are demonstrated in class, and two are yours to complete. Each has a companion Excel calculator.

> **Instructor edition.** This copy ends with an answer key for Decisions 3 and 4. Remove those pages before copying the workbook for students.

How to Use This Workbook

This workbook applies one formula to four business decisions. Decisions 1 and 2 are worked in class as demonstrations, and every step is shown. Decisions 3 and 4 are yours to complete, using the same steps.

Each decision has a companion Excel file that does the arithmetic and draws the chart. Open the file, change the yellow cells, and watch the answer and the graph move. The exercises tell you which file to use.

The Formula You Will Use

Every decision in this workbook is solved by the same rule. It is stated once below.

$$a^* = \mathrm{argmax}_{a \in A} \int U(a, \theta) \cdot p(\theta \mid D)\, d\theta$$

Read it as a sentence. The best action is the one, drawn from the actions available to you, that gives the highest average utility across the possible states of the world, where each state is weighted by how probable it is given the data you hold. The four parts are the action set A, the states theta, the belief p, and the utility U. A fuller treatment is in the lecture note that accompanies this workbook.

The Six Elements Feed the Four Terms

Likierman's six elements of good judgment are the disciplines that supply the formula's four parts. Experience feeds the prior, because a sound starting probability comes from having seen many similar situations. Learning and Trust feed the likelihood, because together they fix how far a piece of evidence should move you. Detachment and Delivery feed the utility, because they keep the scoring free of ego and priced as the outcome will actually unfold. Options feeds the action set, because the argmax can only choose from what you listed. When a decision goes wrong, this mapping is the map of where to look.

Bayes' rule in one line

Your updated belief in a state equals your prior belief multiplied by how well that state explains the evidence, rescaled so the beliefs sum to one. A miniature shows it. Two states start at 0.50 each, and a test arrives that occurs 90 percent of the time under the first and 30 percent under the second. Multiply to get 0.45 and 0.15, then rescale. The first state now carries 0.45 / 0.60 = 0.75. Every update in this workbook is that same arithmetic.

INSTRUCTOR DEMONSTRATION

Decision 1: The Certificate Launch

A school is deciding whether to launch a new professional certificate. Demand is uncertain, and a market survey has just returned a favorable signal. The steps below are worked in full.

Step 1. List the actions. The two actions are to launch the certificate or to hold it.

Step 2. List the states. Demand will be high or low.

Step 3. Set the prior. Before the survey, the school judged high demand at 40 percent and low demand at 60 percent.

Step 4. Set the likelihood. A favorable survey appears 80 percent of the time when demand is high and 30 percent of the time when demand is low.

Step 5. Update to the posterior with Bayes' rule. Multiply prior by likelihood for each state, then rescale to sum to one.

high: $0.80 \times 0.40 = 0.32$ low: $0.30 \times 0.60 = 0.18$ posterior high $= 0.32 / 0.50 = 0.64$

Step 6. Attach the payoffs. Launch and high demand is +500, launch and low demand is −200, and holding is 0, in thousands of dollars.

Step 7. Compute the expected value against the posterior. Holding is zero, and launching is the weighted average of its outcomes.

launch: $0.64 \times 500 + 0.36 \times (-200) = 320 - 72 = 248$

Step 8. Price the risk. Using an exponential utility with a risk tolerance of one million dollars, the certainty equivalent of launching is about 189 thousand.

Step 9. Decide. Launching beats holding on both readings, so the school launches.

Answer. Launch the certificate. The expected value is 248 thousand dollars, and the risk-adjusted value is about 189 thousand, both above holding at zero.

Companion Excel file: Decision-1-Certificate-Launch.xlsx

INSTRUCTOR DEMONSTRATION

Decision 2: The Fixed-Price Bid

A services firm is deciding whether to bid a fixed-price contract of one million dollars. The complexity of the job is unknown. This demonstration shows two lessons. Honest utility can flip a choice, and a third action can beat the first two.

Step 1. List the first actions. The two obvious actions are to bid or to decline.

Step 2. List the states. The job is simple, at a cost of 600 thousand, or complex, at a cost of 1,400 thousand.

Step 3. Set the prior. With no information, each state is 50 percent.

Step 4. Attach the accounting payoffs. Bid and simple is +400, bid and complex is −400, and declining is 0.

Step 5. Compute the naive expected value of bidding. It is a coin flip worth zero, tied with declining.

$$\text{bid: } 0.50 \times 400 + 0.50 \times (-400) = 0$$

Step 6. Correct the utility. A blown complex job carries rework and reputation cost beyond the accounting loss. Value that at 150 thousand, so the complex outcome is −550 and bidding is worth −75.

$$\text{bid, honest: } 0.50 \times 400 + 0.50 \times (-550) = -75$$

Step 7. Enlarge the action set. Add a scoping study that costs 50 thousand and reveals the true complexity before any commitment.

Step 8. Value the study path. Bid only when the job is simple, decline when it is complex, and subtract the study cost.

$$\text{study path: } 0.50 \times 400 + 0.50 \times 0 - 50 = 150$$

Step 9. Check the value of information. Perfect knowledge is worth 200 before cost, so a 50 study sits well under the ceiling.

Step 10. Decide. Run the study, then bid only if the job proves simple.

Answer. Do not bid and do not simply decline. Run the 50 thousand dollar study, then bid only if the job is simple. That path is worth about 150 thousand, which the first two actions never reached.

Companion Excel file: Decision-2-Fixed-Price-Bid.xlsx

Reading the Charts

Every Excel file in this workbook draws a chart, and the same two objects from the formula appear on all of them.

The integral is a height. Every point on a value line or curve is one completed integral, the average of an action's outcomes weighted by the current belief. On the Decision 1 chart it is the height of the value line, which reads 248 at the posterior. On the Decision 2 chart each action is its own line, and the best action at any belief is simply the highest line. On the Decision 3 chart every point on the profit curve is an integral across the whole attendance distribution, computed by the spreadsheet. On the Decision 4 chart the candidates can be counted, so the integral becomes a sum, which is its discrete form.

The derivative is a slope, and the maximum sits where the slope is zero. On the Decision 3 chart the profit curve rises, flattens, and falls, and the best order sits at the flat top, where one more workbook stops paying for itself. The Decision 4 curve has the same shape, and the best cutoff sits where the curve stops rising. On the Decision 1 chart the bend of the risk-adjusted curve below the straight value line is the second derivative of the utility at work, which is risk aversion made visible.

Keep three habits. Height is worth, so compare actions by comparing heights. Slope is sensitivity, so a steep line means a small error in belief moves the answer a long way. Curvature is risk attitude, so a visible bend means the risk adjustment is doing real work. Use these habits in the two exercises that follow.

STUDENT EXERCISE

Decision 3: The Workbook Order

A professional-education unit prints workbooks for a paid workshop. Attendance is uncertain, with a mean of 200 and a standard deviation of 40. Each attendee who receives a workbook earns 60 dollars of margin. Each workbook printed costs 15 dollars. Leftover workbooks have no salvage value.

Your task is to use the formula to find the number of workbooks to print. This decision has a continuous action, because you can print any quantity, and a continuous state, because attendance can take any value. Work through the steps, then use the Excel file to compute and check.

Step 1. Name the action. State what you are choosing and the range it can take.

Step 2. Name the state. State what is uncertain and how its likelihood is described.

Step 3. Write the utility. Write the profit if you print a quantity and attendance turns out to be some number.

Step 4. Find the cost of being one short and the cost of being one over.

Step 5. Compute the critical ratio from those two costs.

Step 6. Read the best order quantity from the critical ratio, using the Excel file.

Step 7. Compare the best order with simply ordering the mean attendance, and explain the difference in one sentence.

Record your answer

Companion Excel file: Decision-3-Workbook-Order.xlsx

STUDENT EXERCISE

Decision 4: The Sequential Hire

A manager is hiring one person. Candidates are interviewed one at a time, in random order. An offer must be made or refused on the spot, and a candidate who is passed over cannot be recalled. There are 50 candidates.

Your task is to use the formula to find the best stopping rule. This decision is sequential, because the choice at each step is to stop or to continue. Work through the steps, then use the Excel file to find the answer.

Step 1. Name the actions at each candidate. State the two choices you face at every step.

Step 2. Name the state. State what you know about the candidate in front of you.

Step 3. Write the utility. State what counts as success and what it is worth.

Step 4. Describe the rule the formula produces. It rejects a first group and then accepts under a condition. State the condition.

Step 5. Use the Excel file to find the best number to screen first and the chance of hiring the best candidate.

Step 6. Express the cutoff as a fraction of the pool, and connect it to the 37 percent rule from the earlier session.

Record your answer

Companion Excel file: Decision-4-Sequential-Hire.xlsx

INSTRUCTOR ANSWER KEY · DECISIONS 3 AND 4

These pages are for the instructor. Remove them before copying the workbook for students.

Answer Key: Decision 3, The Workbook Order

1. The action is the number of workbooks to print, written Q. The action set is continuous, since any whole number is allowed.
2. The state is the actual attendance, written D. It is described by a normal distribution with a mean of 200 and a standard deviation of 40, which is the prior.
3. The utility is the profit. It is 60 dollars for each attendee served, minus 15 dollars for each workbook printed.
4. Being one short costs the lost margin, which is 60 minus 15, or 45 dollars. Being one over costs the print minus the salvage, which is 15 minus 0, or 15 dollars.
5. The critical ratio is the underage cost over the sum of the two costs.

$$\text{critical ratio} = 45 / (45 + 15) = 0.75$$

6. The best order is the point where 75 percent of the demand distribution sits below. That point is 0.674 standard deviations above the mean, so the order is 200 + 0.674 × 40, which rounds to 227.

$$Q^* = 200 + 0.674 \times 40 \approx 227$$

7. The expected profit at 227 is about 8,237 dollars. Ordering the mean of 200 earns about 8,043, so the mean leaves roughly 194 dollars on the table.
8. The best order sits above the mean because being short costs three times as much as being long. Whenever the critical ratio is above one half, the order is above the mean.

Answer. Print about 227 workbooks. In the Excel file the best order is in cell C16 and the expected profit is in C17.

Answer Key: Decision 4, The Sequential Hire

1. At each candidate the two actions are to stop and hire the person in front of you, or to continue.
2. The state is whether the current candidate is the best seen so far. You know only the rank of the current candidate among those already interviewed.
3. The utility is 1 if the person hired is the best of all 50, and 0 otherwise. Maximizing expected utility maximizes the chance of hiring the best.
4. The rule the formula produces is a cutoff. Reject the first r candidates no matter how good, then hire the next candidate who is better than everyone seen so far.

5. For 50 candidates, the best number to screen first is 18, and the chance of hiring the best candidate is about 0.374.

$$r^* = 18 \text{ of } 50 \quad \text{success} \approx 0.374$$

6. The cutoff as a fraction is 18 over 50, which is 0.36. That is close to 1 over e, which is about 0.368. As the pool grows, both the cutoff and the success rate approach 1 over e. This is the 37 percent rule from Session 2.

> **Answer.** Screen the first 18 candidates, then hire the next record-setter. In the Excel file the best cutoff is in cell C8 and the success probability is in C9.